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Programmes and Curriculum — B.Tech IT and M.Tech IT

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Source-type legend used in tables: **O** = official (quoted from a GUNI PDF), **DO** = derived_official (structured/transformed from a GUNI PDF), **S** = synthetic (invented for CampusMate testing). Official page numbers refer to physical PDF pages of DOC-OFFICIAL-SYLLABUS-001 (GUNI structured syllabus) or DOC-OFFICIAL-CATALOG-001 (GUNI 1.1.3 course list).

Semester-wise curriculum of B.Tech IT (PROG-BTECH-IT-001) and M.Tech IT (PROG-MTECH-IT-001). All course codes, titles, credits, hours and marks are transcribed from the official GUNI structured syllabus (Effective Year 2021-2022, dated 28/09/2022) and verified against the printed semester totals. Synthetic content is limited to the labelled programme summaries.

PROG-BTECH-IT-001 — Bachelor of Technology in Information Technology (B.TECH-IT)

source_type: **official (GUNI PDF)** · DOC-OFFICIAL-SYLLABUS-001 p.10

program_id	PROG-BTECH-IT-001
programme name	Bachelor of Technology in Information Technology [O]
short name	B.TECH-IT [O]
degree type	Bachelor of Technology (B.Tech) [DO]
hosted by	COLLEGE-UVPCE-001 U.V. Patel College of Engineering, SCHOOL-FET-001 Faculty of Engineering & Technology [O]
department	DEPT-IT-001 Department of Information Technology [S]
semesters	8 (SEM-BTECH-IT-01, SEM-BTECH-IT-02, SEM-BTECH-IT-03, SEM-BTECH-IT-04, SEM-BTECH-IT-05, SEM-BTECH-IT-06, SEM-BTECH-IT-07, SEM-BTECH-IT-08) [DO]
total credits	174.0 (sum of official semester totals) [DO]
effective year	2021-2022 [O]
catalog program code	202 = BTech(Information Technology) (DOC-OFFICIAL-CATALOG-001) [O]
programme description	Not available in official sources.
synthetic summary	Synthetic summary: an eight-semester undergraduate programme combining engineering foundations (Semesters I–II), core computing (III–IV), specialisation electives and a three-stage capstone project (V–VII) and a full semester of industrial training (VIII). [S]
programme coordinator	FAC-IT-020 [S]

SEM-BTECH-IT-01 — B.TECH-IT Semester I

source_type: **official (GUNI PDF)** · DOC-OFFICIAL-SYLLABUS-001 p.10 · Total Course 7, Regular 6, Elective 0, Open 0, Audit 1

Course ID	Code	Title	Type	TH	PR	Cr	Hrs	Marks	Src
COURSE-2BS101	2BS101	MATHEMATICS - I	Regular	4.0	0.0	4.0	4.0	100	O
COURSE-2ES101	2ES101	ENGINEERING GRAPHICS	Regular	3.0	1.0	4.0	5.0	150	O
COURSE-2ES103	2ES103	BASIC ELECTRICAL ENGINEERING	Regular	3.0	1.0	4.0	5.0	150	O
COURSE-2ES109	2ES109	BASICS OF WEB TECHNOLOGY	Regular	3.0	1.0	4.0	5.0	150	O
COURSE-2ES110	2ES110	IT PERIPHERALS & TOOLS	Regular	0.0	2.0	2.0	4.0	50	O
COURSE-2HS102	2HS102	SEMINAR	Regular	0.0	2.0	2.0	4.0	50	O
EGRP-BTECH-IT-01-AUDIT	—	AUDIT SUBJECT - 1: choose one of 2AC103	Audit	0.0	0.0	0.0	2.0	0	O
		SEMESTER TOTAL (printed in syllabus)		13.0	7.0	20.0	29.0	650	O

Group ID	Group	Option course ID	Code	Course name	Src
EGRP-BTECH-IT-01-AUDIT	AUDIT	COURSE-2AC103	2AC103	ENVIRONMENTAL STUDIES AND DISASTER MANAGEMENT	O

SEM-BTECH-IT-02 — B.TECH-IT Semester II

source_type: **official (GUNI PDF)** · DOC-OFFICIAL-SYLLABUS-001 p.11 · Total Course 8, Regular 6, Elective 1, Open 0, Audit 1

Course ID	Code	Title	Type	TH	PR	Cr	Hrs	Marks	Src
COURSE-2BS102	2BS102	MATHEMATICS - II	Regular	4.0	0.0	4.0	4.0	100	O
COURSE-2BS103	2BS103	PHYSICS	Regular	2.0	1.0	3.0	4.0	150	O
COURSE-2HS101	2HS101	COMMUNICATION SKILLS	Regular	2.0	1.0	3.0	4.0	150	O
COURSE-2ES102	2ES102	WORKSHOP MANUFACTURING PRACTICE	Regular	0.0	2.0	2.0	4.0	50	O
COURSE-2ES104	2ES104	PROGRAMMING FOR PROBLEM SOLVING	Regular	2.0	2.0	4.0	6.0	150	O
COURSE-2ES111	2ES111	IT WORKSHOP	Regular	0.0	2.0	2.0	4.0	50	O
EGRP-BTECH-IT-02-UHV-ELECTIVE-I	—	ELECTIVE SUBJECT - 1 (UHV_Elective-I): choose one of UHV1IPDC1, UHV3DYSBG1	Elective	2.0	0.0	2.0	2.0	100	O
EGRP-BTECH-IT-02-AUDIT	—	AUDIT SUBJECT - 1: choose one of 2AC103	Audit	0.0	0.0	0.0	2.0	0	O
		SEMESTER TOTAL (printed in syllabus)		12.0	8.0	20.0	30.0	750	O

Group ID	Group	Option course ID	Code	Course name	Src
EGRP-BTECH-IT-02-UHV-ELECTIVE-I	UHV_ELECTIVE-I	COURSE-UHV1IPDC1	UHV1IPDC1	UNIVERSAL HUMAN VALUE (INTEGRATED PERSONALITY DEVELOPMENT COURSE-1)	O
EGRP-BTECH-IT-02-UHV-ELECTIVE-I	UHV_ELECTIVE-I	COURSE-UHV3DYSBG1	UHV3DYSBG1	UNIVERSAL HUMAN VALUE (DISCOVER YOURSELF THROUGH BHAGAVAD GITA-1)	O
EGRP-BTECH-IT-02-AUDIT	AUDIT	COURSE-2AC103	2AC103	ENVIRONMENTAL STUDIES AND DISASTER MANAGEMENT	O

SEM-BTECH-IT-03 — B.TECH-IT Semester III

source_type: **official (GUNI PDF)** · DOC-OFFICIAL-SYLLABUS-001 p.12 · Total Course 7, Regular 5, Elective 1, Open 0, Audit 1

Course ID	Code	Title	Type	TH	PR	Cr	Hrs	Marks	Src
COURSE-2BS3102	2BS3102	DISCRETE MATHEMATICS & PROBABILITY	Regular	4.0	0.0	4.0	4.0	100	O
COURSE-2CEIT301	2CEIT301	DIGITAL ELECTRONICS	Regular	3.0	1.0	4.0	5.0	150	O
COURSE-2CEIT302	2CEIT302	OBJECT ORIENTED PROGRAMMING	Regular	3.0	2.0	5.0	7.0	150	O
COURSE-2CEIT303	2CEIT303	DATABASE MANAGEMENT SYSTEM	Regular	3.0	1.0	4.0	5.0	150	O
COURSE-2CEIT304	2CEIT304	DATA STRUCTURES	Regular	3.0	2.0	5.0	7.0	150	O
EGRP-BTECH-IT-03-AUDIT	—	AUDIT SUBJECT - 1: choose one of 2AC301	Audit	0.0	0.0	0.0	2.0	0	O
EGRP-BTECH-IT-03-UHV-ELECTIVE-II	—	ELECTIVE SUBJECT - 1 (UHV_Elective-II): choose one of UHV2IPDC2, UHV4DYSBG2	Elective	2.0	0.0	2.0	2.0	100	O
		SEMESTER TOTAL (printed in syllabus)		18.0	6.0	24.0	32.0	800	O

Group ID	Group	Option course ID	Code	Course name	Src
EGRP-BTECH-IT-03-AUDIT	AUDIT	COURSE-2AC301	2AC301	AUDIT COURSE	O
EGRP-BTECH-IT-03-UHV-ELECTIVE-II	UHV_ELECTIVE-II	COURSE-UHV2IPDC2	UHV2IPDC2	UNIVERSAL HUMAN VALUE (INTEGRATED PERSONALITY DEVELOPMENT COURSE-2)	O

Group ID	Group	Option course ID	Code	Course name	Src
EGRP-BTECH-IT-03-UHV-ELECTIVE-II	UHV_ELECTIVE-II	COURSE-UHV4DYSBG2	UHV4DYSBG2	UNIVERSAL HUMAN VALUE (DISCOVER YOURSELF THROUGH BHAGAVAD GITA-2)	O

SEM-BTECH-IT-04 — B.TECH-IT Semester IV

source_type: **official (GUNI PDF)** · DOC-OFFICIAL-SYLLABUS-001 p.13 · Total Course 6, Regular 6, Elective 0, Open 0, Audit 0

Course ID	Code	Title	Type	TH	PR	Cr	Hrs	Marks	Src
COURSE-2BS4101	2BS4101	MATHEMATICS FOR COMPUTER ENGINEERING & INFORMATION TECHNOLOGY	Regular	4.0	0.0	4.0	4.0	100	O
COURSE-2CEIT401	2CEIT401	OPERATING SYSTEMS	Regular	3.0	1.0	4.0	5.0	150	O
COURSE-2CEIT402	2CEIT402	DESIGN AND ANALYSIS OF ALGORITHMS	Regular	3.0	1.0	4.0	5.0	150	O
COURSE-2CEIT403	2CEIT403	APPLICATION DEVELOPMENT TOOLS	Regular	2.0	2.0	4.0	6.0	150	O
COURSE-2CEIT404	2CEIT404	PROGRAMMING WITH PYTHON	Regular	3.0	1.0	4.0	5.0	150	O
COURSE-2CEIT405	2CEIT405	NOSQL DATABASE SYSTEMS	Regular	0.0	1.0	1.0	2.0	50	O
		SEMESTER TOTAL (printed in syllabus)		15.0	6.0	21.0	27.0	750	O

SEM-BTECH-IT-05 — B.TECH-IT Semester V

source_type: **official (GUNI PDF)** · DOC-OFFICIAL-SYLLABUS-001 p.14 · Total Course 7, Regular 5, Elective 2, Open 0, Audit 0

Course ID	Code	Title	Type	TH	PR	Cr	Hrs	Marks	Src
COURSE-2CEIT501	2CEIT501	COMPUTER ARCHITECTURE & ORGANIZATION	Regular	3.0	1.0	4.0	5.0	150	O
COURSE-2CEIT502	2CEIT502	SOFTWARE ENGINEERING	Regular	3.0	1.0	4.0	5.0	150	O
COURSE-2CEIT503	2CEIT503	COMPUTER NETWORKS	Regular	4.0	1.0	5.0	6.0	150	O
COURSE-2CEIT504	2CEIT504	CAPSTONE PROJECT-I	Regular	0.0	1.0	1.0	2.0	50	O
COURSE-2HS5101	2HS5101	APTITUDE SKILL BUILDING-I	Regular	0.0	1.0	1.0	2.0	50	O
EGRP-BTECH-IT-05-ELECTIVE-I	—	ELECTIVE SUBJECT - 1 (ELECTIVE-I): choose one of 2CEIT5PE1, 2CEIT5PE2, 2CEIT5PE3, 2CEIT5PE4	Elective	3.0	1.0	4.0	5.0	150	O
EGRP-BTECH-IT-05-ELECTIVE-II	—	ELECTIVE SUBJECT - 1 (ELECTIVE-II): choose one of 2CEIT5PE5, 2CEIT5PE6, 2CEIT5PE7, 2CEIT5PE8	Elective	2.0	2.0	4.0	6.0	150	O
		SEMESTER TOTAL (printed in syllabus)		15.0	8.0	23.0	31.0	850	O

Group ID	Group	Option course ID	Code	Course name	Src
EGRP-BTECH-IT-05-ELECTIVE-I	ELECTIVE-I	COURSE-2CEIT5PE1	2CEIT5PE1	ADVANCED JAVA	O
EGRP-BTECH-IT-05-ELECTIVE-I	ELECTIVE-I	COURSE-2CEIT5PE2	2CEIT5PE2	COMPUTER GRAPHICS & VISUALIZATION	O
EGRP-BTECH-IT-05-ELECTIVE-I	ELECTIVE-I	COURSE-2CEIT5PE3	2CEIT5PE3	QUANTUM COMPUTING	O
EGRP-BTECH-IT-05-ELECTIVE-I	ELECTIVE-I	COURSE-2CEIT5PE4	2CEIT5PE4	SOFTWARE PACKAGES	O
EGRP-BTECH-IT-05-ELECTIVE-II	ELECTIVE-II	COURSE-2CEIT5PE5	2CEIT5PE5	MOBILE APPLICATION DEVELOPMENT	O
EGRP-BTECH-IT-05-ELECTIVE-II	ELECTIVE-II	COURSE-2CEIT5PE6	2CEIT5PE6	HUMAN COMPUTER INTERACTION	O
EGRP-BTECH-IT-05-ELECTIVE-II	ELECTIVE-II	COURSE-2CEIT5PE7	2CEIT5PE7	INNOVATION & ENTREPRENEURSHIP	O
EGRP-BTECH-IT-05-ELECTIVE-II	ELECTIVE-II	COURSE-2CEIT5PE8	2CEIT5PE8	ADVANCED PROGRAMMING WITH .NET	O

SEM-BTECH-IT-06 — B.TECH-IT Semester VI

source_type: **official (GUNI PDF)** · DOC-OFFICIAL-SYLLABUS-001 p.15 · Total Course 8, Regular 6, Elective 2, Open 0, Audit 0

Course ID	Code	Title	Type	TH	PR	Cr	Hrs	Marks	Src
COURSE-2CEIT601	2CEIT601	THEORY OF COMPUTATION	Regular	3.0	1.0	4.0	5.0	150	O
COURSE-2CEIT602	2CEIT602	ARTIFICIAL INTELLIGENCE	Regular	3.0	1.0	4.0	5.0	150	O
COURSE-2CEIT603	2CEIT603	CLOUD COMPUTING	Regular	3.0	1.0	4.0	5.0	150	O
COURSE-2CEIT604	2CEIT604	USER EXPERIENCE DESIGN	Regular	0.0	1.0	1.0	2.0	50	O
COURSE-2CEIT605	2CEIT605	CAPSTONE PROJECT-II	Regular	0.0	1.0	1.0	2.0	50	O
COURSE-2HS6101	2HS6101	APTITUDE SKILL BUILDING–II	Regular	0.0	1.0	1.0	2.0	50	O
EGRP-BTECH-IT-06-ELECTIVE-III	—	ELECTIVE SUBJECT - 1 (ELECTIVE-III): choose one of 2CEIT6PE1, 2CEIT6PE2, 2CEIT6PE3, 2CEIT6PE4, 2CEIT6PE5	Elective	3.0	1.0	4.0	5.0	150	O
EGRP-BTECH-IT-06-ELECTIVE-IV	—	ELECTIVE SUBJECT - 1 (ELECTIVE-IV): choose one of 2CEIT6PE6, 2CEIT6PE7, 2CEIT6PE8, 2CEIT6PE9	Elective	3.0	1.0	4.0	5.0	150	O
		SEMESTER TOTAL (printed in syllabus)		15.0	8.0	23.0	31.0	900	O

Group ID	Group	Option course ID	Code	Course name	Src
EGRP-BTECH-IT-06-ELECTIVE-III	ELECTIVE-III	COURSE-2CEIT6PE1	2CEIT6PE1	WEB TECHNOLOGY	O
EGRP-BTECH-IT-06-ELECTIVE-III	ELECTIVE-III	COURSE-2CEIT6PE2	2CEIT6PE2	SOFTWARE VALIDATION & TESTING	O
EGRP-BTECH-IT-06-ELECTIVE-III	ELECTIVE-III	COURSE-2CEIT6PE3	2CEIT6PE3	CRYPTOGRAPHY AND NETWORK SECURITY	O
EGRP-BTECH-IT-06-ELECTIVE-III	ELECTIVE-III	COURSE-2CEIT6PE4	2CEIT6PE4	DATA SCIENCES AND MACHINE LEARNING	O
EGRP-BTECH-IT-06-ELECTIVE-III	ELECTIVE-III	COURSE-2CEIT6PE5	2CEIT6PE5	GAME DESIGN & DEVELOPMENT	O
EGRP-BTECH-IT-06-ELECTIVE-IV	ELECTIVE-IV	COURSE-2CEIT6PE6	2CEIT6PE6	ROBOTICS	O
EGRP-BTECH-IT-06-ELECTIVE-IV	ELECTIVE-IV	COURSE-2CEIT6PE7	2CEIT6PE7	ETHICAL HACKING	O
EGRP-BTECH-IT-06-ELECTIVE-IV	ELECTIVE-IV	COURSE-2CEIT6PE8	2CEIT6PE8	IT INFRASTRUCTURE & MANAGEMENT	O
EGRP-BTECH-IT-06-ELECTIVE-IV	ELECTIVE-IV	COURSE-2CEIT6PE9	2CEIT6PE9	INTERNET OF THINGS	O

SEM-BTECH-IT-07 — B.TECH-IT Semester VII

source_type: **official (GUNI PDF)** · DOC-OFFICIAL-SYLLABUS-001 p.16, 17 · Total Course 6, Regular 4, Elective 2, Open 0, Audit 0

Course ID	Code	Title	Type	TH	PR	Cr	Hrs	Marks	Src
COURSE-2CEIT701	2CEIT701	COMPILER DESIGN	Regular	3.0	1.0	4.0	5.0	150	O
COURSE-2CEIT702	2CEIT702	BIG DATA ANALYTICS	Regular	3.0	1.0	4.0	5.0	150	O
COURSE-2CEIT703	2CEIT703	CAPSTONE PROJECT-III	Regular	0.0	6.0	6.0	12.0	100	O
COURSE-2CEIT704	2CEIT704	SOFT SKILLS	Regular	0.0	1.0	1.0	2.0	50	O
EGRP-BTECH-IT-07-ELECTIVE-V	—	ELECTIVE SUBJECT - 1 (ELECTIVE-V): choose one of 2CEIT78PE1, 2CEIT78PE2, 2CEIT78PE3, 2CEIT78PE4, 2CEIT78PE5, 2CEIT78PE6	Elective	3.0	1.0	4.0	5.0	150	O
EGRP-BTECH-IT-07-ELECTIVE-VI	—	ELECTIVE SUBJECT - 1 (ELECTIVE-VI): choose one of 2CEIT78PE7, 2CEIT78PE8, 2CEIT78PE9, 2CEIT78PE10, 2CEIT78PE11, 2CEIT78PE12, 2CEIT78PE13	Elective	3.0	1.0	4.0	5.0	150	O
		SEMESTER TOTAL (printed in syllabus)		12.0	11.0	23.0	34.0	750	O

Group ID	Group	Option course ID	Code	Course name	Src
EGRP-BTECH-IT-07-ELECTIVE-V	ELECTIVE-V	COURSE-2CEIT78PE1	2CEIT78PE1	DEEP LEARNING	O
EGRP-BTECH-IT-07-ELECTIVE-V	ELECTIVE-V	COURSE-2CEIT78PE2	2CEIT78PE2	BLOCKCHAIN TECHNOLOGIES	O
EGRP-BTECH-IT-07-ELECTIVE-V	ELECTIVE-V	COURSE-2CEIT78PE3	2CEIT78PE3	FUNDAMENTALS OF IMAGE PROCESSING	O
EGRP-BTECH-IT-07-ELECTIVE-V	ELECTIVE-V	COURSE-2CEIT78PE4	2CEIT78PE4	MOBILE COMPUTING	O
EGRP-BTECH-IT-07-ELECTIVE-V	ELECTIVE-V	COURSE-2CEIT78PE5	2CEIT78PE5	SEARCH ENGINE OPTIMIZATION	O
EGRP-BTECH-IT-07-ELECTIVE-V	ELECTIVE-V	COURSE-2CEIT78PE6	2CEIT78PE6	VIRTUAL REALITY	O
EGRP-BTECH-IT-07-ELECTIVE-VI	ELECTIVE-VI	COURSE-2CEIT78PE7	2CEIT78PE7	NATURAL LANGUAGE PROCESSING	O

Group ID	Group	Option course ID	Code	Course name	Src
EGRP-BTECH-IT-07-ELECTIVE-VI	ELECTIVE-VI	COURSE-2CEIT78PE8	2CEIT78PE8	FORENSICS & CYBER LAW	O
EGRP-BTECH-IT-07-ELECTIVE-VI	ELECTIVE-VI	COURSE-2CEIT78PE9	2CEIT78PE9	ADVANCED ALGORITHMS	O
EGRP-BTECH-IT-07-ELECTIVE-VI	ELECTIVE-VI	COURSE-2CEIT78PE10	2CEIT78PE10	FOG COMPUTING	O
EGRP-BTECH-IT-07-ELECTIVE-VI	ELECTIVE-VI	COURSE-2CEIT78PE11	2CEIT78PE11	FAULT TOLERANT SYSTEMS	O
EGRP-BTECH-IT-07-ELECTIVE-VI	ELECTIVE-VI	COURSE-2CEIT78PE12	2CEIT78PE12	HIGH PERFORMANCE COMPUTING	O
EGRP-BTECH-IT-07-ELECTIVE-VI	ELECTIVE-VI	COURSE-2CEIT78PE13	2CEIT78PE13	COMPUTER VISION	O

SEM-BTECH-IT-08 — B.TECH-IT Semester VIII

source_type: **official (GUNI PDF)** · DOC-OFFICIAL-SYLLABUS-001 p.18 · Total Course 1, Regular 1, Elective 0, Open 0, Audit 0

Course ID	Code	Title	Type	TH	PR	Cr	Hrs	Marks	Src
COURSE-2CEIT801	2CEIT801	INDUSTRIAL TRAINING	Regular	0.0	20.0	20.0	40.0	400	O
		SEMESTER TOTAL (printed in syllabus)		0.0	20.0	20.0	40.0	400	O

PROG-MTECH-IT-001 — Master of Technology in Information Technology (M.TECH-IT)

source_type: **official (GUNI PDF)** · DOC-OFFICIAL-SYLLABUS-001 p.104

program_id	PROG-MTECH-IT-001
programme name	Master of Technology in Information Technology [O]
short name	M.TECH-IT [O]
degree type	Master of Technology (M.Tech) [DO]
hosted by	COLLEGE-UVPCE-001 U.V. Patel College of Engineering, SCHOOL-FET-001 Faculty of Engineering & Technology [O]
department	DEPT-IT-001 Department of Information Technology [S]
semesters	4 (SEM-MTECH-IT-01, SEM-MTECH-IT-02, SEM-MTECH-IT-03, SEM-MTECH-IT-04) [DO]
total credits	74.0 (sum of official semester totals) [DO]
effective year	2021-2022 [O]
catalog program code	402 = Mtech(Information Technology) (DOC-OFFICIAL-CATALOG-001) [O]
programme description	Not available in official sources.
synthetic summary	Synthetic summary: a four-semester postgraduate programme with advanced core courses and electives in Semesters I–II followed by a two-part major project and seminar in Semesters III–IV. [S]
programme coordinator	FAC-IT-013 [S]

SEM-MTECH-IT-01 — M.TECH-IT Semester I

source_type: **official (GUNI PDF)** · DOC-OFFICIAL-SYLLABUS-001 p.104 · Total Course 6, Regular 4, Elective 2, Open 0, Audit 0

Course ID	Code	Title	Type	TH	PR	Cr	Hrs	Marks	Src
COURSE-3CEIT101	3CEIT101	COMPUTATIONAL INTELLIGENCE	Regular	3.0	1.0	4.0	5.0	150	O
COURSE-3CEIT102	3CEIT102	COMPUTER ALGORITHMS	Regular	3.0	1.0	4.0	5.0	150	O
COURSE-3CEIT103	3CEIT103	CRYPTOGRAPHY AND NETWORK SECURITY	Regular	3.0	1.0	4.0	5.0	150	O
COURSE-3CEIT104	3CEIT104	RESEARCH ORIENTATION	Regular	0.0	1.0	1.0	2.0	50	O
EGRP-MTECH-IT-01-ELECTIVE-I	—	ELECTIVE SUBJECT - 1 (ELECTIVE-I): choose one of 3CEIT105, 3CEIT106, 3CEIT107, 3CEIT108, 3CEIT109	Elective	3.0	1.0	4.0	5.0	150	O
EGRP-MTECH-IT-01-ELECTIVE-II	—	ELECTIVE SUBJECT - 1 (ELECTIVE-II): choose one of 3CEIT110, 3CEIT111, 3CEIT112, 3CEIT113	Elective	3.0	1.0	4.0	5.0	150	O
		SEMESTER TOTAL (printed in syllabus)		15.0	6.0	21.0	27.0	800	O

Group ID	Group	Option course ID	Code	Course name	Src
EGRP-MTECH-IT-01-ELECTIVE-I	ELECTIVE-I	COURSE-3CEIT105	3CEIT105	WIRELESS NETWORKS	O
EGRP-MTECH-IT-01-ELECTIVE-I	ELECTIVE-I	COURSE-3CEIT106	3CEIT106	ADVANCED TOPICS IN NETWORKS	O
EGRP-MTECH-IT-01-ELECTIVE-I	ELECTIVE-I	COURSE-3CEIT107	3CEIT107	INTERNET TECHNOLOGIES	O
EGRP-MTECH-IT-01-ELECTIVE-I	ELECTIVE-I	COURSE-3CEIT108	3CEIT108	NETWORK PROGRAMMING AND MANAGEMENT	O
EGRP-MTECH-IT-01-ELECTIVE-I	ELECTIVE-I	COURSE-3CEIT109	3CEIT109	HIGH PERFORMANCE COMPUTING	O
EGRP-MTECH-IT-01-ELECTIVE-II	ELECTIVE-II	COURSE-3CEIT110	3CEIT110	DISTRIBUTED COMPUTING	O
EGRP-MTECH-IT-01-ELECTIVE-II	ELECTIVE-II	COURSE-3CEIT111	3CEIT111	SERVICE ORIENTED COMPUTING	O
EGRP-MTECH-IT-01-ELECTIVE-II	ELECTIVE-II	COURSE-3CEIT112	3CEIT112	ADVANCE SOFTWARE ENGINEERING	O
EGRP-MTECH-IT-01-ELECTIVE-II	ELECTIVE-II	COURSE-3CEIT113	3CEIT113	REAL TIME SYSTEMS	O

SEM-MTECH-IT-02 — M.TECH-IT Semester II

source_type: **official (GUNI PDF)** · DOC-OFFICIAL-SYLLABUS-001 p.105, 106 · Total Course 12, Regular 3, Elective 8, Open 1, Audit 0

Course ID	Code	Title	Type	TH	PR	Cr	Hrs	Marks	Src
COURSE-3CEIT201	3CEIT201	DIGITAL IMAGE PROCESSING	Regular	3.0	1.0	4.0	5.0	150	O
COURSE-3CEIT202	3CEIT202	DATA MINING AND DATA WAREHOUSING	Regular	3.0	1.0	4.0	5.0	150	O
COURSE-3CEIT203	3CEIT203	CLOUD COMPUTING	Regular	3.0	1.0	4.0	5.0	150	O
EGRP-MTECH-IT-02-ELECTIVE-III	—	ELECTIVE SUBJECT - 1 (ELECTIVE-III): choose one of 3CEIT204, 3CEIT205, 3CEIT206, 3CEIT207	Elective	3.0	1.0	4.0	5.0	150	O
EGRP-MTECH-IT-02-ELECTIVE-IV	—	ELECTIVE SUBJECT - 1 (ELECTIVE-IV): choose one of 3CEIT208, 3CEIT209, 3CEIT210, 3CEIT211	Elective	3.0	1.0	4.0	5.0	150	O
EGRP-MTECH-IT-02-OPEN	—	OPEN SUBJECT - 1: choose one of 3OS201	Open	3.0	0.0	3.0	3.0	100	O
		SEMESTER TOTAL (printed in syllabus)		18.0	5.0	23.0	28.0	850	O

Group ID	Group	Option course ID	Code	Course name	Src
EGRP-MTECH-IT-02-ELECTIVE-III	ELECTIVE-III	COURSE-3CEIT204	3CEIT204	ADVANCED OPERATING SYSTEM	O
EGRP-MTECH-IT-02-ELECTIVE-III	ELECTIVE-III	COURSE-3CEIT205	3CEIT205	ADVANCED COMPUTER ARCHITECTURE	O
EGRP-MTECH-IT-02-ELECTIVE-III	ELECTIVE-III	COURSE-3CEIT206	3CEIT206	SOFTWARE ARCHITECTURE AND DESIGN PATTERN	O
EGRP-MTECH-IT-02-ELECTIVE-III	ELECTIVE-III	COURSE-3CEIT207	3CEIT207	PUBLIC KEY INFRASTRUCTURE	O
EGRP-MTECH-IT-02-ELECTIVE-IV	ELECTIVE-IV	COURSE-3CEIT208	3CEIT208	SATELLITE NETWORKS	O
EGRP-MTECH-IT-02-ELECTIVE-IV	ELECTIVE-IV	COURSE-3CEIT209	3CEIT209	ESSENTIAL MATHEMATICS	O
EGRP-MTECH-IT-02-ELECTIVE-IV	ELECTIVE-IV	COURSE-3CEIT210	3CEIT210	MODELING AND SIMULATION	O
EGRP-MTECH-IT-02-ELECTIVE-IV	ELECTIVE-IV	COURSE-3CEIT211	3CEIT211	ADVANCED DATABASE MANAGEMENT SYSTEM	O
EGRP-MTECH-IT-02-OPEN	OPEN	COURSE-3OS201	3OS201	OPEN SUBJECT	O

SEM-MTECH-IT-03 — M.TECH-IT Semester III

source_type: **official (GUNI PDF)** · DOC-OFFICIAL-SYLLABUS-001 p.107 · Total Course 4, Regular 2, Elective 2, Open 0, Audit 0

Course ID	Code	Title	Type	TH	PR	Cr	Hrs	Marks	Src
COURSE-3CEIT301	3CEIT301	MAJOR PROJECT PART - I	Regular	0.0	8.0	8.0	16.0	200	O
COURSE-3CEIT302	3CEIT302	SEMINAR	Regular	0.0	2.0	2.0	4.0	100	O
EGRP-MTECH-IT-03-ELECTIVE-V	—	ELECTIVE SUBJECT - 1 (ELECTIVE-V): choose one of 3CEIT303, 3CEIT304, 3CEIT305, 3CEIT306	Elective	3.0	1.0	4.0	5.0	150	O
EGRP-MTECH-IT-03-ELECTIVE-VI	—	ELECTIVE SUBJECT - 1 (ELECTIVE-VI): choose one of 3CEIT307, 3CEIT308, 3CEIT309, 3CEIT310	Elective	3.0	1.0	4.0	5.0	150	O
		SEMESTER TOTAL (printed in syllabus)		6.0	12.0	18.0	30.0	600	O

Group ID	Group	Option course ID	Code	Course name	Src
EGRP-MTECH-IT-03-ELECTIVE-V	ELECTIVE-V	COURSE-3CEIT303	3CEIT303	SEMANTIC WEB AND SOCIAL NETWORKS	O
EGRP-MTECH-IT-03-ELECTIVE-V	ELECTIVE-V	COURSE-3CEIT304	3CEIT304	DATA COMPRESSION	O
EGRP-MTECH-IT-03-ELECTIVE-V	ELECTIVE-V	COURSE-3CEIT305	3CEIT305	INTERNET OF THINGS	O
EGRP-MTECH-IT-03-ELECTIVE-V	ELECTIVE-V	COURSE-3CEIT306	3CEIT306	BIG DATA ANALYTICS	O
EGRP-MTECH-IT-03-ELECTIVE-VI	ELECTIVE-VI	COURSE-3CEIT307	3CEIT307	NATURAL LANGUAGE PROCESSING	O
EGRP-MTECH-IT-03-ELECTIVE-VI	ELECTIVE-VI	COURSE-3CEIT308	3CEIT308	COMPILER DESIGN	O
EGRP-MTECH-IT-03-ELECTIVE-VI	ELECTIVE-VI	COURSE-3CEIT309	3CEIT309	DEEP LEARNING	O
EGRP-MTECH-IT-03-ELECTIVE-VI	ELECTIVE-VI	COURSE-3CEIT310	3CEIT310	CYBER SECURITY	O

SEM-MTECH-IT-04 — M.TECH-IT Semester IV

source_type: **official (GUNI PDF)** · DOC-OFFICIAL-SYLLABUS-001 p.108 · Total Course 1, Regular 1, Elective 0, Open 0, Audit 0

Course ID	Code	Title	Type	TH	PR	Cr	Hrs	Marks	Src
COURSE-3CEIT401	3CEIT401	MAJOR PROJECT PART - II	Regular	0.0	12.0	12.0	24.0	300	O
		SEMESTER TOTAL (printed in syllabus)		0.0	12.0	12.0	24.0	300	O

Examination grade range & value (official)

source_type: **official (GUNI PDF)** · DOC-OFFICIAL-SYLLABUS-001 p.10 (repeated on every semester page)

Grade band ID	Marks/Perc. range	Grade (Regular)	Grade point (Regular)	Grade (Remedial)	Grade point (Remedial)
GRADE-BAND-01	90>=	A+	10	A	9
GRADE-BAND-02	80 To 89	A	9	B+	8
GRADE-BAND-03	70 To 79	B+	8	B	7
GRADE-BAND-04	60 To 69	B	7	C+	6
GRADE-BAND-05	50 To 59	C+	6	C	5
GRADE-BAND-06	40 To 49	C	5	C	5
GRADE-BAND-07	<40	F	0	F	0

Notes on interpretation

source_type: **derived_official (structured from GUNI PDF)**

• The examination scheme prints marks as '0/40', '0/60' etc. CampusMate stores the raw string and interprets the denominator as maximum marks (derived_official).

• Elective, audit and open slots carry one credit/marks scheme per slot. Each option course inherits that slot's scheme (derived_official).

• Prerequisites, course descriptions, teaching departments and faculty are NOT given in the official syllabus pages. Any such values in CampusMate are synthetic and labelled.